```
!pip -q install spacy
!python -m spacy download en_core_web_sm -q

import spacy
import pandas as pd
from collections import Counter

nlp = spacy.load("en_core_web_sm")
```

```
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import pandas as pd

# Create a sample DataFrame with a 'Text' column for testing
data = {
    "Text": [
        "Natural language processing (NLP) is a machine learning technology that gives computers t
    ]
}

df = pd.DataFrame(data)
display(df)
```

	Text
0	Natural language processing (NLP) is a machine...

```
text = str(df["Text"].iloc[0])

doc = nlp(text[:5000])

print("Original Text:")
print(text[:500])

print("\nTokens:")
print([token.text for token in doc if not token.is_space])

print("\nLemmas:")
print([token.lemma_ for token in doc if not token.is_space])

print("\nParts of Speech:")
print([(token.text, token.pos_) for token in doc if not token.is_space])
```

```
Original Text:
Natural language processing (NLP) is a machine learning technology that gives computers the abili

Tokens:
['Natural', 'language', 'processing', '(', 'NLP', ')', 'is', 'a', 'machine', 'learning', 'technol

Lemmas:
['natural', 'language', 'processing', '(', 'NLP', ')', 'be', 'a', 'machine', 'learn', 'technology

Parts of Speech:
[('Natural', 'ADJ'), ('language', 'NOUN'), ('processing', 'NOUN'), ('(', 'PUNCT'), ('NLP', 'PROPN
```

```
print("Nouns:")
print([token.text for token in doc if token.pos_ == "NOUN"])
```

```
print("\nVerbs:")
print([token.text for token in doc if token.pos_ == "VERB"])

print("\nAdjectives:")
print([token.text for token in doc if token.pos_ == "ADJ"])

print("\nAdverbs:")
print([token.text for token in doc if token.pos_ == "ADV"])
```

```
Nouns:
['language', 'processing', 'machine', 'technology', 'computers', 'ability', 'manipulate', 'langua

Verbs:
['learning', 'gives', 'interpret', 'understand', 'makes', 'tokenize', 'extract', 'find']

Adjectives:
['Natural', 'human', 'excellent', 'easy']

Adverbs:
['efficiently']
```

```
import matplotlib.pyplot as plt

pos_counts = Counter(
    token.pos_
    for token in doc
    if not token.is_space
)

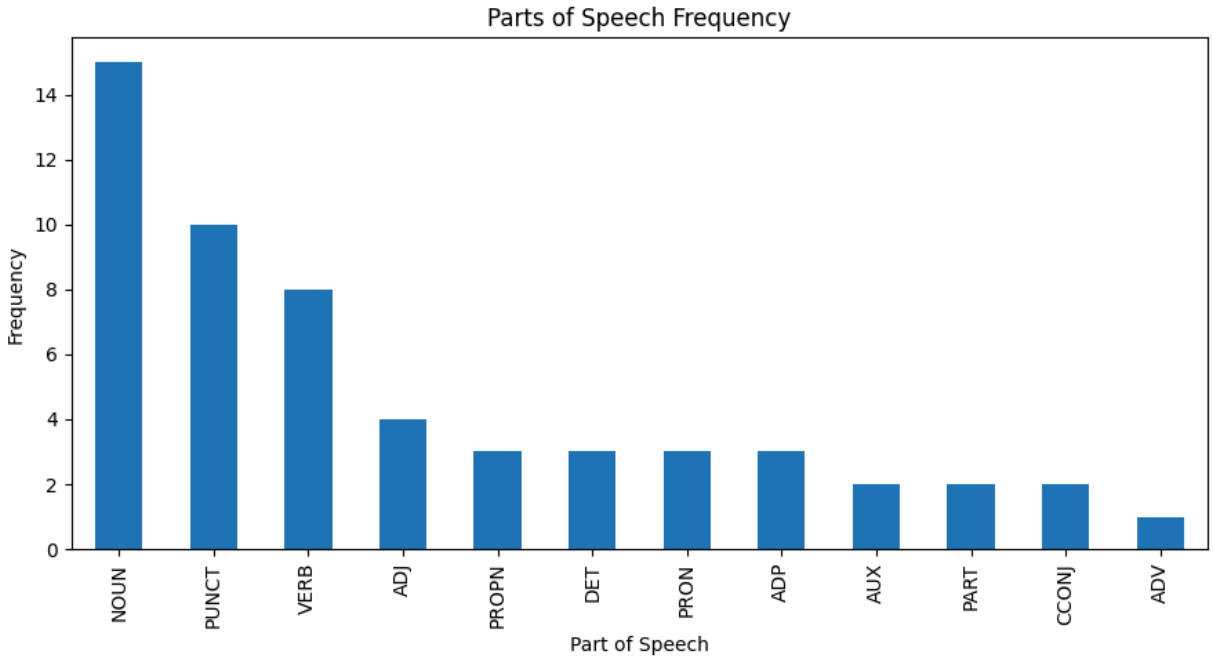
pos_df = pd.DataFrame(
    pos_counts.items(),
    columns=["Part of Speech", "Count"]
).sort_values("Count", ascending=False)

display(pos_df)

pos_df.plot(
    x="Part of Speech",
    y="Count",
    kind="bar",
    legend=False,
    figsize=(9, 5)
)

plt.title("Parts of Speech Frequency")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```

	Part of Speech	Count
1	NOUN	15
2	PUNCT	10
6	VERB	8
0	ADJ	4
3	PROPN	3
5	DET	3
7	PRON	3
10	ADP	3
4	AUX	2
8	PART	2
9	CCONJ	2
11	ADV	1



Next steps: Generate code with pos_df | New interactive sheet

```
print("Named Entities:")

for entity in doc.ents:
    print(entity.text, "->", entity.label_)
```

```
Named Entities:
NLP -> ORG
Spacy -> PERSON
NLP -> ORG
```

```
sample_texts = [
    "Apple launched a new product in California in 2025.",
    "John bought coffee for 15 dollars in New York.",
    "The company delivered the product on Monday."
]

for text in sample_texts:
    doc = nlp(text)
```

```
    print("\nText:", text)

    for entity in doc.ents:
        print(entity.text, "->", entity.label_)
```

```
Text: Apple launched a new product in California in 2025.
Apple -> ORG
California -> GPE
2025 -> DATE

Text: John bought coffee for 15 dollars in New York.
John -> PERSON
15 dollars -> MONEY
New York -> GPE

Text: The company delivered the product on Monday.
Monday -> DATE
```